



OSCE-Project: Generative Adversarial Agents System for evaluating Objective Structured Clinical Examination capabilities

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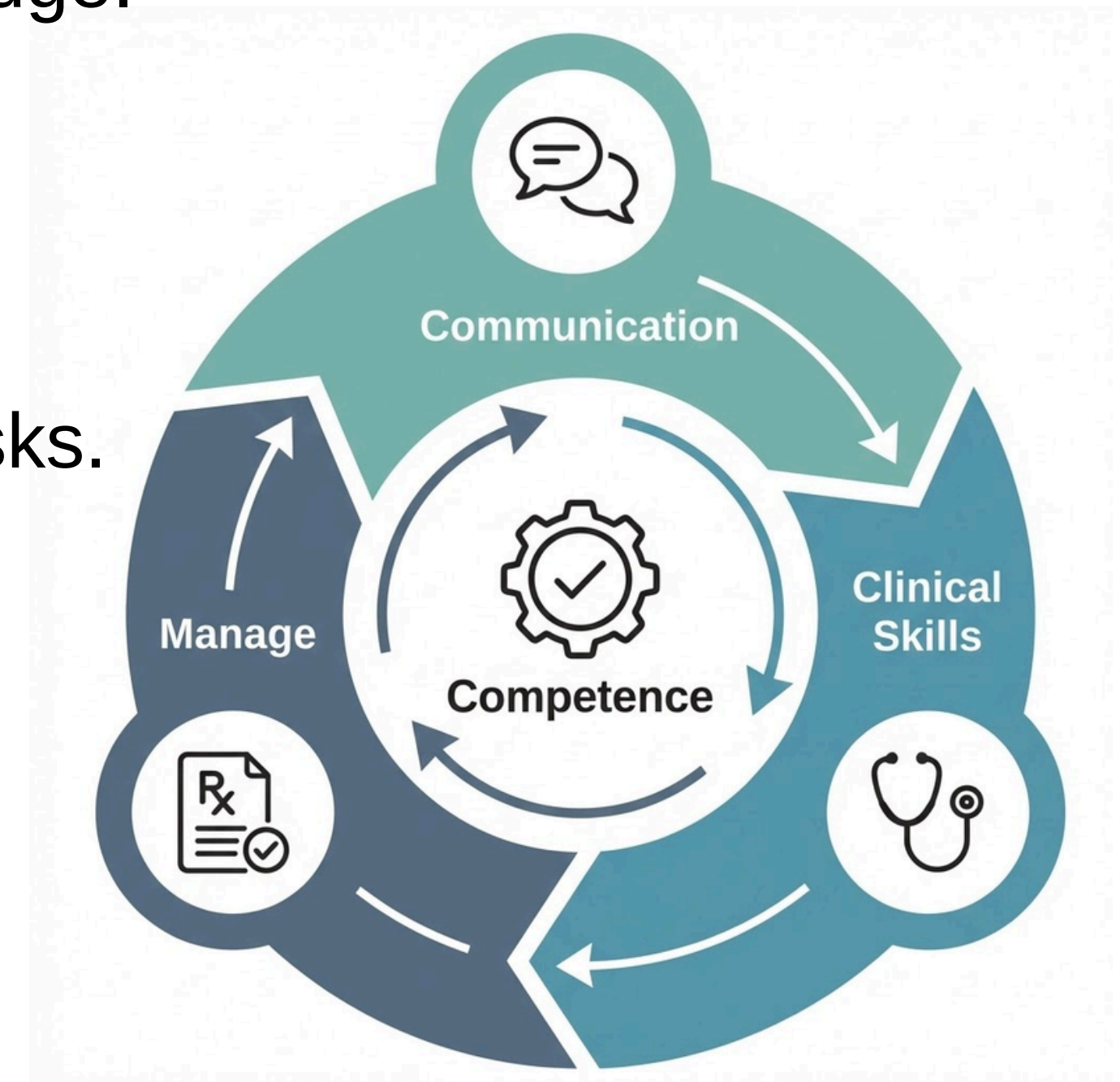


Slides

What is an OSCE?

Objective Structured Clinical Examination (OSCE) Benchmark for assessing "shows how" competence, beyond textbook knowledge.

- **Objective:** Same stations & criteria for all.
- **Structured:** Specific, time-bound, real-world tasks.
- **Core Competencies:**
 - **Communication** (History & empathy).
 - **Clinical Skills** (Exams & procedures).
 - **Management** (Decisions & prescriptions).



The "Standardized Patient" Bottleneck



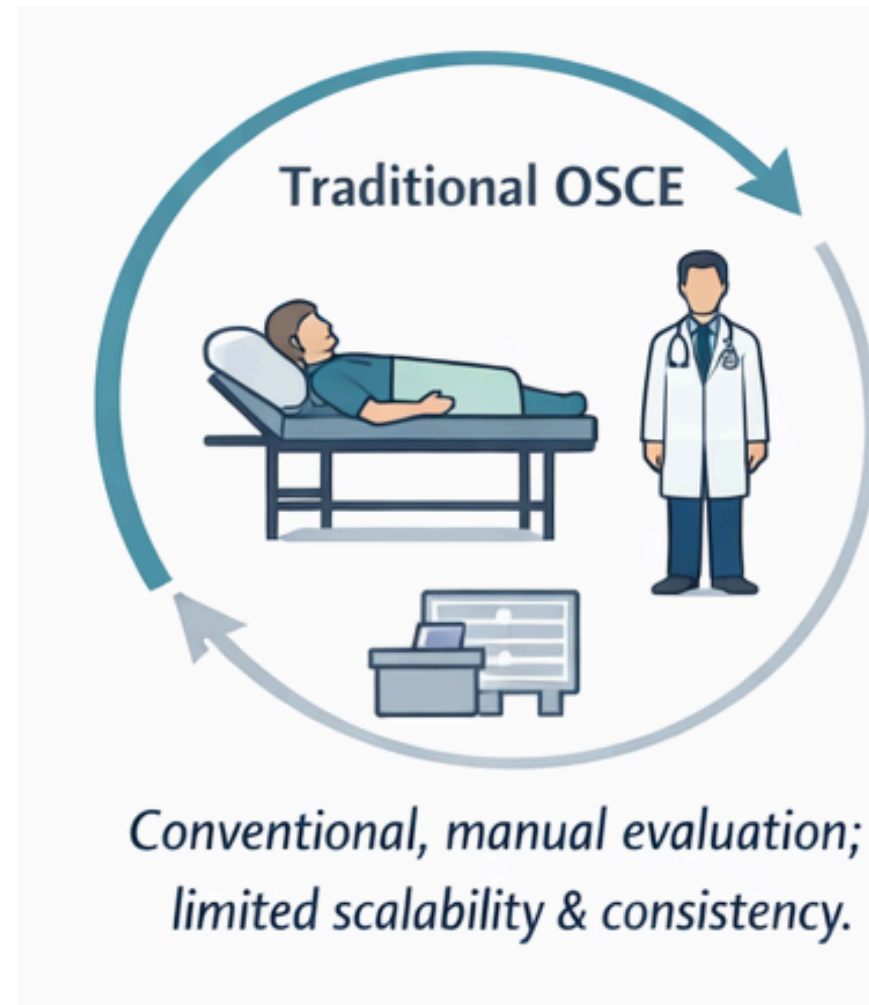
Diversity Limitations
in Traditional OSCEs



Costly exams and tired actor
cause inconsistency.

Our Solution: The GAA Framework

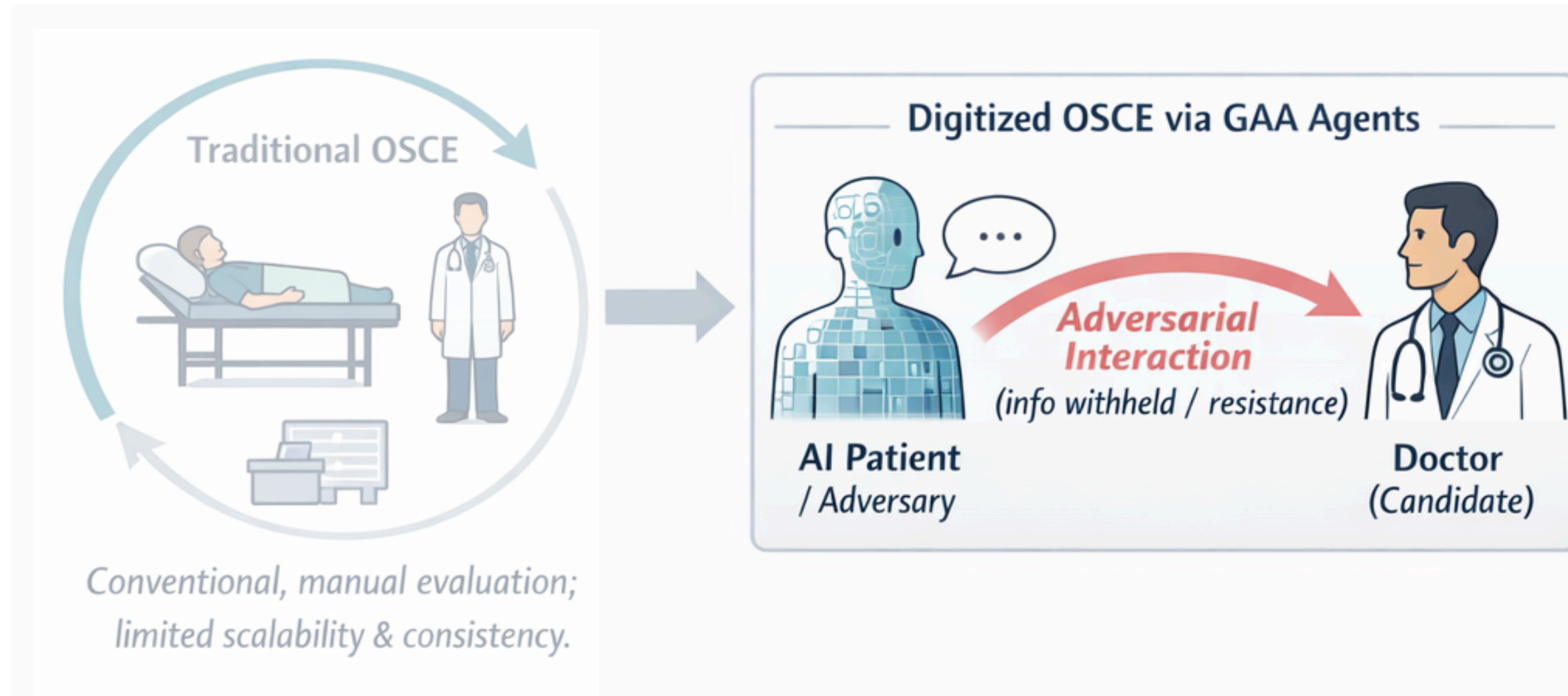
Generative Adversarial Agents for Automatic Evaluation



Traditional exams rely on **expensive actors** and **subjective grading**, which severely limits student practice opportunities.

Our Solution: The GAA Framework

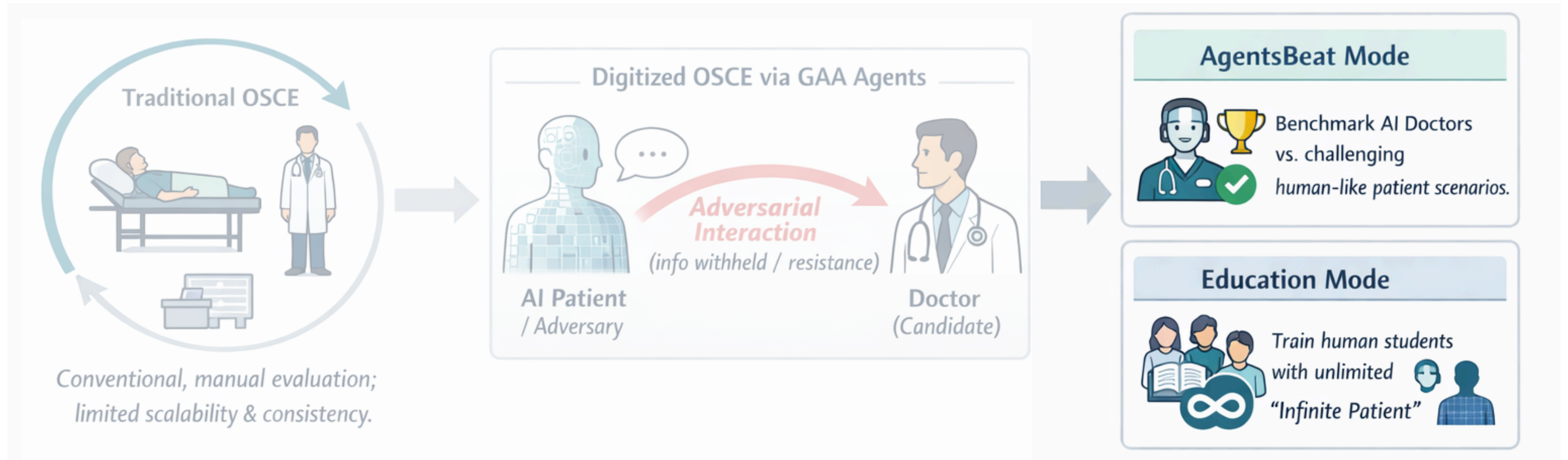
Generative Adversarial Agents for Automatic Evaluation



We digitize the exam using **Adversarial AI Patients** that actively withhold information to stress-test the doctor's communication skills.

Our Solution: The GAA Framework

Generative Adversarial Agents for Automatic Evaluation



This system powers two critical modes: a rigorous **benchmark for AI models** and an **'Infinite Patient' simulator** for human training.

Our Solution: The GAA Framework

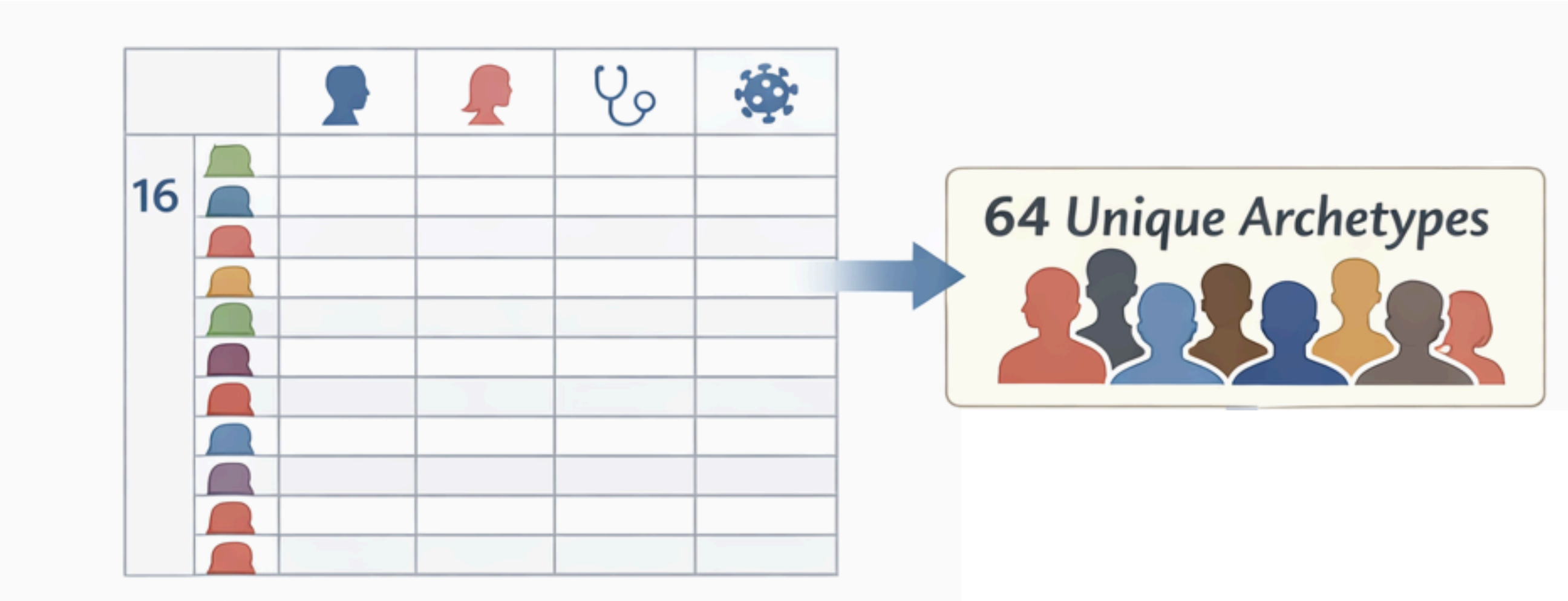
Generative Adversarial Agents for Automatic Evaluation



The result is a **human-centric evaluation framework** that is scalable, consistent, and free from human bias.

How to Constructing "Diverse" Patients?

- **Mechanized Diversity:** Generates **64 unique patient** archetypes by mixing personality, gender, and disease profiles.



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- **Mechanized Diversity:** Generates **64 unique patient** archetypes by mixing personality, gender, and disease profiles.
- **Hidden Internal State:** Patient agents run a continuous '**Hidden Monologue**' that logically drives their behavior.



Think (Hidden)

*"The doctor seems dismissive.
I don't trust them."*



Act (Visible)

"Crosses arms and looks away."

System filters out "Think"—Doctor only sees Act/Say



Act (Visible)

"Crosses arms and looks away."



Say (Visible)

"I'm not sure about this treatment."

How to Constructing "Diverse" Patients?

- **Mechanized Diversity:** Generates **64 unique patient** archetypes by mixing personality, gender, and disease profiles.
- **Hidden Internal State:** Patient agents run a continuous '**Hidden Monologue**' that logically drives their behavior.
- **Information Asymmetry:** Doctors must interpret **visible cues ('Act' & 'Say')** to **uncover** the patient's **hidden thoughts**.



Consistent Evaluation Criteria

- **3-Dimensional Scoring:** Scores **30 specific soft-skills** across Empathy, Persuasion, and Safety.



Empathy

Emotional
Connection

*Did the doctor
validate the
patient's fear?*



Persuasion

Communication
Efficacy

*Did the doctor
explain risks
in plain language?*



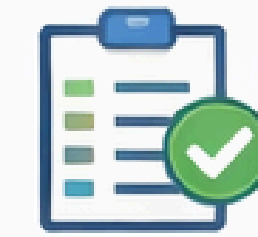
Safety

Clinical
Correctness

*Did the doctor
check for
drug allergies?*

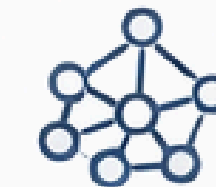
Consistent Evaluation Criteria

- **3-Dimensional Scoring:** Scores **30 specific soft-skills** across Empathy, Persuasion, and Safety.
- **Real-Time Judging:** **Real-time analysis** detects if the patient accepts or refuses care.



Per-Round Judging

- Real-time scoring for each conversation round; trajectories evaluated live

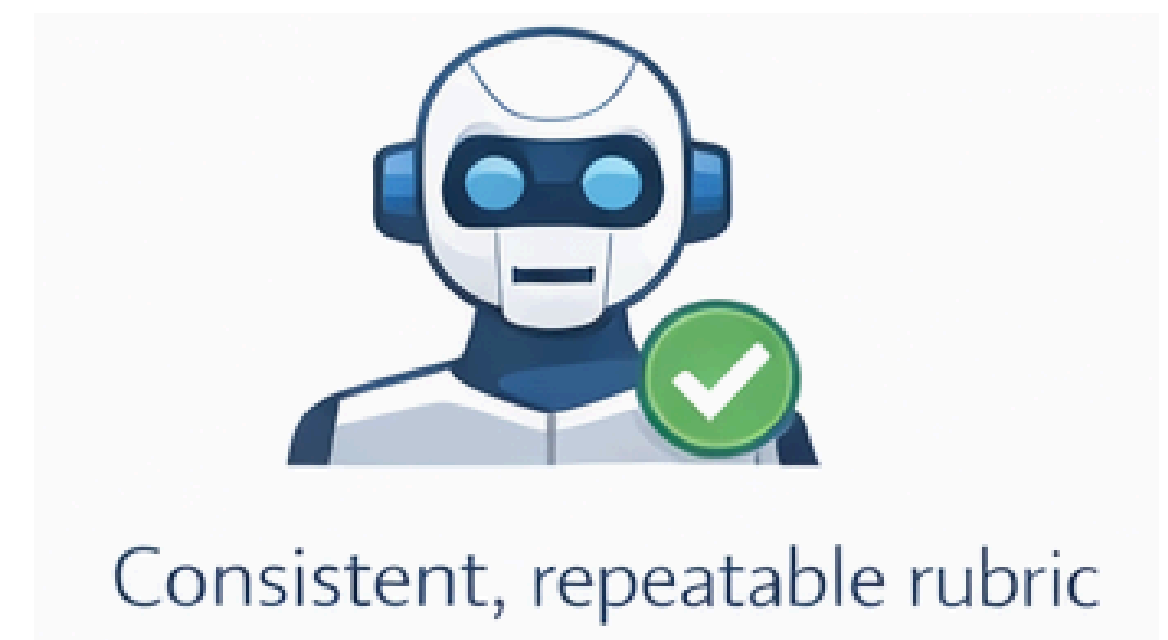
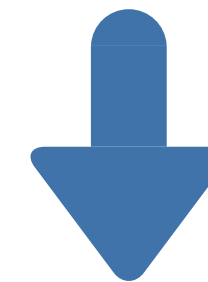


Stop Detector

- Autonomous module terminates exam on:
- **Success:** *Patient Accepted Treatment* ✓
- **Failure:** *Patient Left / Refused Care* ✗

Consistent Evaluation Criteria

- **3-Dimensional Scoring:** Scores **30 specific soft-skills** across Empathy, Persuasion, and Safety.
- **Real-Time Judging:** **Real-time analysis** detects if the patient accepts or refuses care.
- **Eliminating Bias:** Replaces subjective human grading with a **consistent, bias-free AI rubric**.



Conclusion & Future Work

Summary:

- We replaced the "Standardized Patient" bottleneck with a **Generative Patient Engine**.
- We replaced subjective grading with a **Standardized AI Judge**.



Slides

Future Impact:

- **Human Training:** Extending this system to medical schools for **"Infinite Practice"** loops.
- **Multimodal:** Adding voice (ASR) and avatar visuals for **immersive realism**.
- **Global Standard:** A universal benchmark for evaluating both Human and AI doctors.



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